

MELBIN VARGHESE

AI/ML Engineer | Data Science Intern

+91 7558836481 | melbinvarghese77@gmail.com | LinkedIn: melbin-varghese

GitHub: Melbin-Varghese

Kochi, Kerala, India

PROFILE

AI/ML Engineer with hands-on experience in machine learning, computer vision, predictive analytics, and AI-powered application development. Experienced in building end-to-end ML workflows including data preprocessing, feature engineering, model training, evaluation, and deployment using **Python, Scikit-learn, TensorFlow, OpenCV, SQL, and Streamlit**. Familiar with local **LLM workflows, RAG concepts, and practical AI solutions** for real-world automation problems.

SKILLS

- **Programming:** Python, SQL
- **Machine Learning & AI:** Supervised Learning, Unsupervised Learning, Deep Learning, NLP, Computer Vision, Predictive Analytics, RAG
- **Libraries & Frameworks:** Scikit-learn, TensorFlow, OpenCV, MediaPipe, Hugging Face, Pandas, NumPy
- **Data Visualization:** Power BI, Matplotlib, Seaborn
- **Databases & Tools:** MySQL, Git, Jupyter Notebook, VS Code
- **AI/LLM Tools:** Ollama, Embeddings, Vector Databases (basic), ChromaDB

EXPERIENCE

Data Science Intern

2025 – Present

Luminar Technolab | Kochi, Kerala, India

- Built and optimized **ML pipelines for preprocessing, feature engineering**, regression, and classification tasks using Python and Scikit-learn.
- Developed **machine learning models for regression and classification** using Scikit-learn, improving prediction accuracy by **15%**.
- Built **real-time computer vision applications** using YOLO, OpenCV, and MediaPipe for object and gesture detection.
- Built AI-powered applications using **Generative AI**, Large Language Models (LLMs), LangChain, and Ollama for intelligent automation.
- Worked on **deep learning, predictive analytics, and model evaluation** for AI-based applications.

PROJECTS

AI-Based Hand Gesture Recognition Game

Apr 2026

Project Link

- Developed a **real-time hand gesture recognition system** using **OpenCV, MediaPipe, and CVZone**.
- Implemented gesture detection and tracking pipeline for computer vision-based interaction.
- Integrated **text-to-speech (TTS)** and audio feedback for dynamic gameplay experience.
- Optimized OpenCV inference pipeline for smooth real-time performance and low-latency detection.

Machine Learning-Based Traffic Volume Prediction System

Feb 2026

Project Link

- Processed and analyzed **48K+ records** using feature engineering and encoding techniques.
- Developed predictive machine learning models improving prediction accuracy by **15%**.
- Built interactive dashboards improving decision-making speed by **20%**.

Product Expiry Reminder System

Sep 2025

Project Link

- Developed a full-stack inventory management system using **Python, Streamlit, and MySQL** to manage **100+ products**.
- Built automated **email alert system** for expiry tracking, reducing product waste by **40%**.
- Implemented secure authentication and optimized MySQL queries for efficient data handling.

Local LLM-Based Document Assistant using Ollama

Feb 2026

- Built a **Retrieval-Augmented Generation (RAG)** system using **Ollama, LangChain, and ChromaDB** for intelligent document question answering.
- Implemented PDF ingestion, text chunking, embedding generation, and semantic retrieval pipelines.
- Integrated local LLMs and embedding models for privacy-focused document interaction.
- Developed an end-to-end workflow for document preprocessing, vector storage, retrieval, and context-aware response generation.

EDUCATION

B.Tech in Computer Science and Engineering

APJ Abdul Kalam Technological University

2021 – 2025

CGPA: 7.2

CERTIFICATIONS

- **Machine Learning Algorithms** – 365 Data Science

- **GenAI Data Analytics Simulation** – Forage
- **Data Cleaning with Pandas** – 365 Data Science